2022. M81A

2022L562A1EL

Coimisiún na Scrúduithe Stáit

State Examinations Commission

Leaving Certificate Examination, 2022

Design & Communication Graphics

Higher Level

Section A (60 marks)

Centre No.

Thursday, 23 June
Morning, 9:30 - 12:30

This examination is divided into three sections:

SECTION A	(Core - Short Questions)
SECTION B	(Core - Long Questions)
SECTION C	(Applied Graphics - Long Questions)

SECTION A

- Four questions are presented.
- Answer **any three** on the A3 sheet overleaf.
- All questions in Section A carry **20 marks** each.

SECTION B and SECTION C

- Eight questions are presented.
- Answer **any two** on drawing paper.
- All questions in Section B and Section C carry **60 marks** each.

General Instructions:

- *Construction lines must be shown on all solutions.*
- *Write the question number distinctly on the answer paper in Sections B and C.*
- *Work on one side of the drawing paper only.*
- *All dimensions are given in metres or millimetres.*
- *Write your Examination number in the box below and on all other sheets used.*

Examination Number

The 2022 examination papers were adjusted to compensate for disruptions to learning due to COVID-19. This examination paper does not necessarily reflect the same structure and format as the examination papers of past or subsequent years.

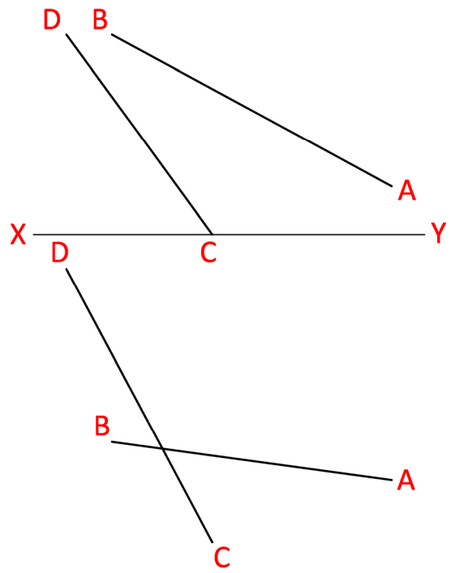
SECTION A - Core - Answer **any three** of the questions on this A3 sheet.

A-1. The image below shows the booms of two cranes. The crane booms are represented by the skew lines **AB** and **CD** on the right.

(a) Determine the projections of the shortest horizontal line between the skew lines.

(b) Determine, and indicate in millimetres, the true length of this line.



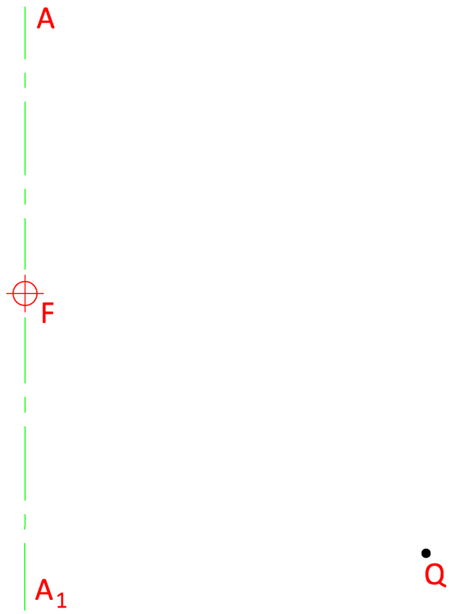


A-2. The image below shows the parabolic bridge at Mizen Head in Cork.
The drawing below shows the axis **AA$_1$**, focus **F**, and two points **P** and **Q** on a similar parabola.

(a) Locate the directrix and the vertex and draw a portion of the parabola which shall pass through **P** and **Q**.

(b) Draw a tangent to the curve at point **Q**.



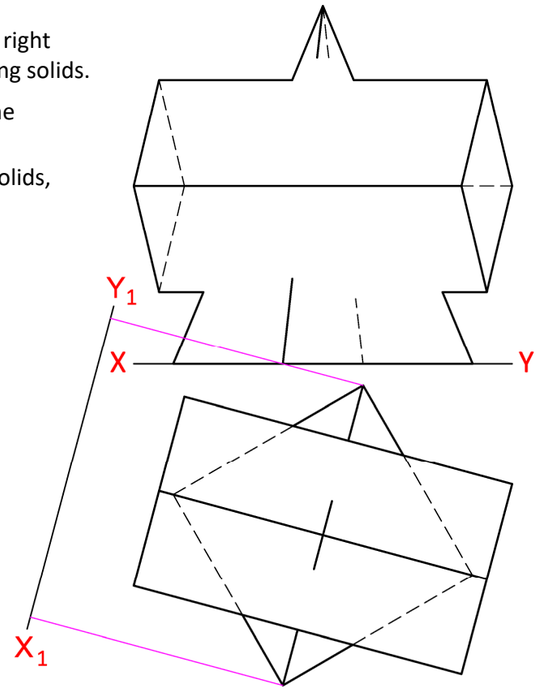


A-3. The image below shows a design based on a prism intersecting a square-based pyramid. The drawing on the right shows the incomplete elevation and plan of the intersecting solids.

(a) Using the given **X$_1$Y$_1$**, draw an auxiliary elevation of the intersecting solids.

(b) Complete the elevation and plan of the intersecting solids, showing all lines of interpenetration.





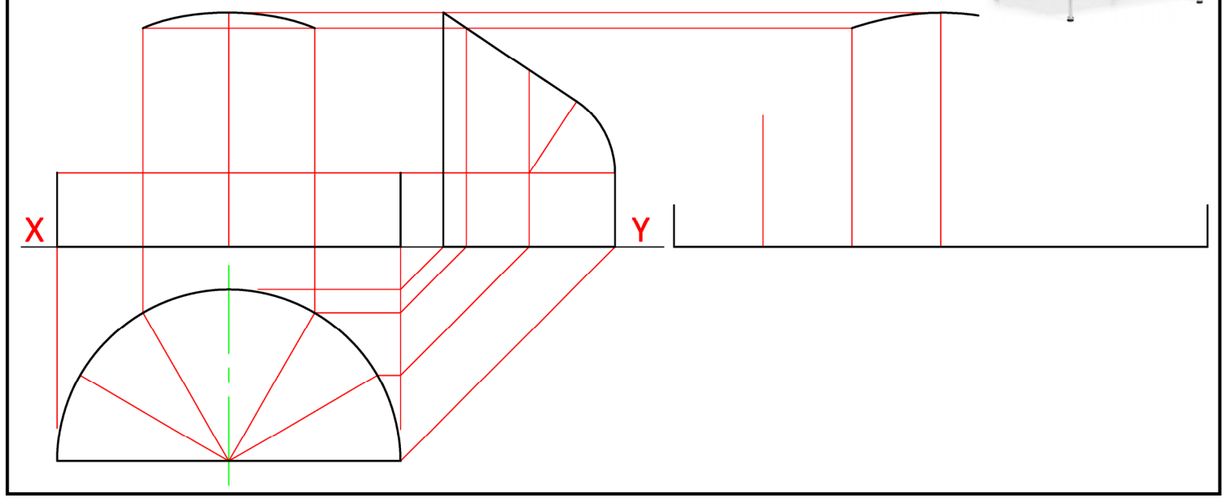
A-4. The image on the right shows a modern chair design. It includes a curved plywood back, based on a cylinder which is truncated and shaped as shown.

The drawing shows the plan, incomplete elevation and end view of the plywood back of a similar chair. The partially completed surface development of the back has been projected from the end view.

(a) Complete the elevation.

(b) Complete the surface development.





This examination paper must be returned at the end of the examination – You must include your Examination Number on the front cover.

This Contour Map is part of Section C and should only be used for the answering of the Geologic Geometry Option (Question C-1).

(Scale 1:1000)

